An AI-Powered Interactive Suspense Role-Playing Game

AIDM7040 Group Project Report

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1 Introduction: Problem and Topic Analysis

Traditional tabletop role-playing games (TRPGs) rely heavily on a human game master (GM), who is responsible for narration, non-player character (NPC) performance, rule explanation, dice judgment, and story progression. This structure gives TRPG a strong sense of freedom and immersion, but it also creates a relatively high entry barrier.

A complete TRPG session usually requires preparation, rule understanding, and an experienced host. For beginners, the complexity of the rules and the lack of guidance can make participation difficult. As a result, although TRPG offers rich interactive storytelling, it is not always easy to access in a lightweight digital form.

To address this problem, our project develops an AI TRPG Bot based on a large language model. In our demo, the system takes over part of the GM's role by narrating scenes, responding as NPCs, performing dice-based checks, and tracking player state.

2 Why Generative AI is the Right Tool

Generative AI is suitable for this project because a TRPG system must do more than present story content. It must also respond to player input in a flexible and meaningful way. This is difficult to achieve with a purely fixed branching structure.

Large language models are well suited to this task because they support open-ended natural language interaction. Prior research has discussed the use of LLMs in games as non-player characters, assistants, and game masters (Gallotta et al., 2024; Kelly et al., 2023). This matches the needs of our project, in which the AI is used to narrate scenes, respond as NPCs, and adapt the story to player actions.

However, generative AI should not be used alone. Pure text generation may reduce consistency

in rules and game state. For this reason, our project adopts a hybrid design: the LLM handles narration and dialogue, while rule-based logic manages dice checks and state tracking. This makes the system more flexible than a fully scripted game and more stable than an unconstrained chatbot.

3 Approach and Methods

3.1 Narrative and Script Design

To make the demo more immersive and coherent, we created an original suspense script for the project. The story is set in the abandoned St. Alden Residential Academy, where the player investigates a series of student disappearances and hidden records. The narrative gradually reveals that the school was involved in illegal hormone experiments in cooperation with an external institution, and that the player is not only an investigator but also one of the surviving "successful samples" from the past. The story includes key characters such as Ethan, Nina, and Lucas, who each support the mystery structure in different ways.

The script was designed with both narrative pacing and gameplay interaction in mind. Instead of writing a fully linear story, we divided the experience into major investigation stages, such as entering the abandoned campus, exploring the Student Wellness Center, discovering hidden underground spaces, revealing the player's identity, and escaping with evidence. We also designed multiple character roles, including detective, hacker, and priest, so that players could enter the same story world with slightly different perspectives and interaction styles.

3.2 Implementation

The overall architecture follows a hybrid design in fig 1. On the front end, the website presents narrative text, player input, and visible game information such as role, danger level, and key status updates. On the back end, the system manages story state and interaction flow. The LLM is mainly responsible for generating scene narration, NPC replies, and action feedback, while rule-based logic is used to keep the game consistent and playable.

A core mechanic of the project is the dice-check system. When the player performs an important action, such as observing clues, persuading a character, or resisting psychological pressure, the system may trigger a roll check. Each check is linked to a skill category, such as observation, persuasion, or willpower. The final result is determined by combining the dice outcome with role-related modifiers, and the result then affects the narrative response and state update. In this way, the game keeps some of the uncertainty and role-playing structure of traditional TRPGs.

The system also includes state-switching logic to control progression. During play, variables such as danger level, health condition, collected clues, and key story flags are continuously updated. State changes happen when the player reaches important scenes, succeeds or fails at

critical checks, or obtains core evidence. These changes determine whether the story moves to a new area, reveals hidden information, triggers identity-related memories, or enters the final escape stage. Therefore, the website is not only a presentation layer, but also a structured game system that connects authored story design, AI generation, and interactive mechanics.

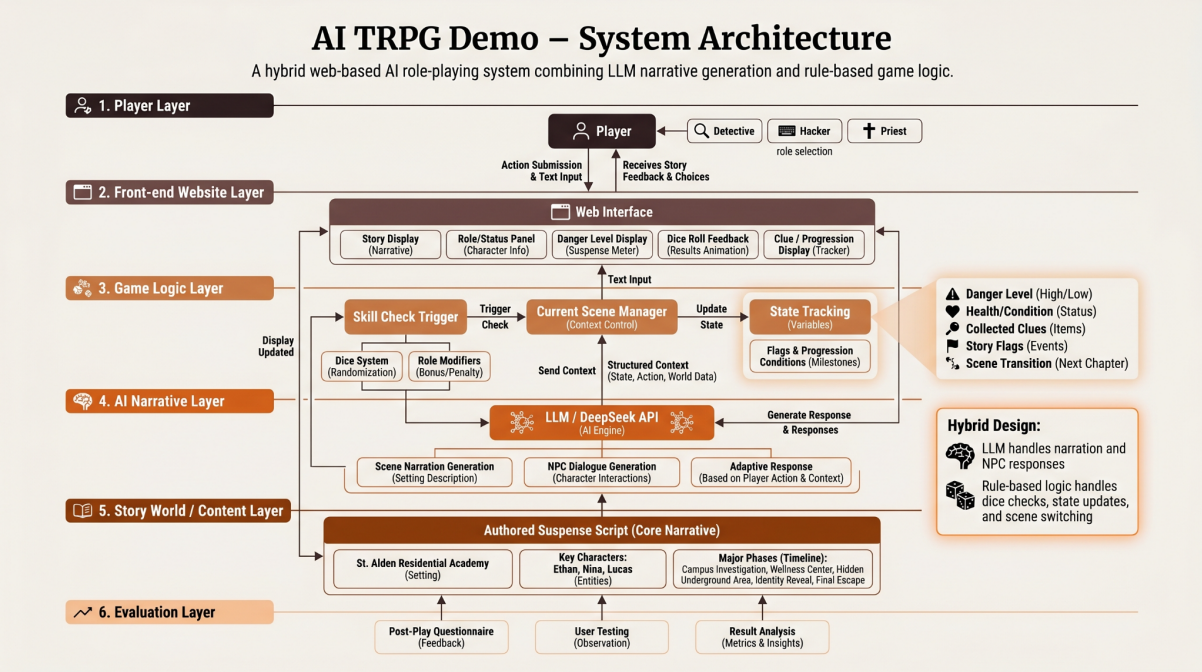

Figure 1: Overall architecture

3.3 Evaluation Method

To evaluate the effectiveness of the demo, we conducted a user test with a post-experience questionnaire. A total of 26 valid responses were collected. The questionnaire included background questions about users' gaming habits and prior exposure to interactive narrative, as well as a series of Likert-scale questions on immersion, plot attraction, interactivity, AI performance, creative stimulation, and overall experience. It also included open-ended questions for qualitative feedback. The full questionnaire is included in Appendix A.

This evaluation method was chosen because our project is not only a technical system but also an interactive narrative experience. Therefore, its quality should not be judged only by whether the system runs correctly, but also by whether users can understand the process, feel engaged by the story, and perceive their choices as meaningful. The questionnaire allowed us to examine both the practical usability of the website and the narrative effectiveness of the AI TRPG format. In addition, the open-ended responses helped us identify specific areas for improvement, such as visual presentation, suspense intensity, pacing of clues, and interaction feedback.

Table 1: Summary of questionnaire results

Dimension / Item	Score	Interpretation
Immersion	3.90 / 5	Users could enter the setting, but suspense atmosphere can be stronger
Plot attraction	4.08 / 5	The opening and mystery structure were effective
Interactivity	4.08 / 5	Players generally felt their choices mattered
AI system performance	4.12 / 5	AI responses were relevant and coherent overall
Creativity stimulation	3.95 / 5	The demo inspired ideas, but this effect was moderate
Overall evaluation	4.26 / 5	General feedback was positive
Willingness to continue playing	4.35 / 5	The project showed strong replay potential
Willingness to recommend	4.46 / 5	Word-of-mouth potential was very strong
Overall experience rating	8.27 / 10	Most users gave a high overall score

4 Results and Discussion

A total of 26 valid questionnaire responses were collected after the playtest. Overall, the results suggest that the AI TRPG demo achieved a positive user response in terms of story attraction, interactivity, and AI performance. Table 1 summarizes the main findings.

The results indicate three main strengths. First, the narrative design was effective: plot attraction received 4.08/5, and users responded positively to the opening, suspense, and clue structure. Second, interactivity was successfully established. The interactivity dimension also reached 4.08/5, and users generally agreed that their choices were meaningful. Third, the AI component performed well in practice. AI system performance reached 4.12/5, suggesting that most players felt the generated responses matched their input and maintained narrative coherence. These findings support the core design goal of the project, which is to use generative AI to provide a lightweight but still engaging TRPG experience.

At the same time, the questionnaire also revealed several limitations. Immersion (3.90/5) and creativity stimulation (3.95/5) were slightly lower than the other core dimensions, and the detailed responses showed that suspense intensity, step clarity, and visual presentation still need improvement. Open-ended feedback further suggested that the current version contains too much text, limited visual dynamics, and relatively weak feedback in clue exploration and puzzle solving. Therefore, while the demo already demonstrates the practical value of an AI-assisted TRPG system, future improvement should focus on stronger audiovisual immersion, clearer interaction guidance, and more visible response to player actions.

5 Conclusion

This project developed an AI TRPG demo that explores how generative AI can support lightweight role-playing interaction in a digital suspense narrative. By combining LLM-generated narration with rule-based mechanics such as dice checks, state tracking, and scene switching, the system was able to take over part of the traditional game master's role in a more accessible format.

The results suggest that this hybrid design is effective. Users responded positively to the plot, interactivity, and AI performance, and many showed willingness to continue playing and recommend the demo to others. :contentReference[oaicite:2]index=2 At the same time, the current version still needs improvement in suspense intensity, visual presentation, and interaction feedback. :contentReference[oaicite:3]index=3

Overall, the project shows that generative AI can be a practical tool for interactive storytelling when it is combined with structured game logic. It also demonstrates the potential of AI-based role-playing systems as a digital media application that reduces the entry barrier of traditional TRPG.

6 Workload Distribution

Member	Main contribution
ZHANG Yufei	Frontend implementation, session UI, game page interaction, story image integration
HU Xinyuan	Story writing, scenario design, scene flow planning
WU Jiaqian	Questionnaire design, user testing, result analysis, video generate
YIN Hanyu	Report writing, editing, slide organization, project coordination

References

Gallotta, R., Todd, G., Zammit, M., Earle, S., Liapis, A., Togelius, J., & Yannakakis, G. N. (2024). Large language models and games: A survey and roadmap. *arXiv preprint arXiv:2402.18659*.

Kelly, J., Mateas, M., & Wardrip-Fruin, N. (2023). Towards computational support with language models for ttrpg game masters. *Proceedings of the 18th International Conference on the Foundations of Digital Games*.

A Appendix A. Website of the demo

https://ai-trpg-nu.vercel.app/

B Appendix B. User Testing Questionnaire

This questionnaire was used for user testing, result analysis, and report writing in the course project.

Questionnaire Introduction

This questionnaire was designed to understand players' experience after trying the AI TRPG Demo, including ease of understanding, sense of participation, plot attraction, AI system performance, and overall playability. Participants were asked to answer based on their actual experience.

TRPG (Tabletop Role-Playing Game) is a form of game centered on storytelling, role-playing, and interactive decision-making. Players usually take on specific roles, explore scenes, investigate clues, make choices, and push the story forward. The tested work in this project is a TRPG demo that combines AI-driven narration with suspense investigation.

Part I: Background Information and Testing Condition

1. Do you usually play games?
    - Often
    - Sometimes
    - Rarely
    - Almost never
2. Have you experienced any of the following types of content before? (Multiple choice)
    - Suspense genre
    - Interactive narrative games
    - Text adventure games
    - TRPG / tabletop role-playing
    - AI story / AI games
    - Rarely experienced any of the above
3. How long did your playtest take this time?
    - Less than 5 minutes
    - 5–10 minutes
    - 10–15 minutes
    - 15–20 minutes
    - More than 20 minutes

4. Did you complete the main flow of the current demo?
    - Yes, basically completed
    - Partially completed
    - No, I quit halfway

Part II: Experience Evaluation

The following items were rated on a 5-point Likert scale, where 1 = strongly disagree and 5 = strongly agree.

Immersion

5. I could quickly enter the game setting and story atmosphere.
6. The scenes, narration, and atmosphere gave me a sense of immersion.
7. I felt a strong sense of suspense or tension during the playtest.

Plot Attraction

8. The beginning of the game caught my interest.
9. The suspense, clues, and setting made me want to know the truth.
10. The plot progression was logical and easy to follow.

Interactivity

11. I could feel that my choices or actions affected the experience.
12. The game was not just "reading text"; it gave me a sense of participation.
13. I think the choices I made were meaningful.

AI System Performance

14. The AI's responses generally matched my input or choices.
15. The plot progression was coherent before and after each interaction.
16. I could understand what to do at each step.

Creativity Stimulation

17. This demo stimulated my imagination about how the story might develop.
18. It made it easier for me to think of roles, clues, or branch design.
19. After the playtest, I became more willing to try designing an interactive story by myself.
20. This demo helped me understand how suspense narrative progresses.

Overall Evaluation

21. Overall, this playtest experience was interesting.
22. If a more complete version is released later, I would like to continue playing.
23. If it is further improved, I would recommend it to classmates or friends.
24. Please rate the overall experience of this demo on a scale from 1 to 10.

Part III: Open-ended Questions

25. Which part of the demo did you like most?
26. What do you think most needs improvement at present?
27. Did you encounter any difficulty in understanding, operation, or pacing during the playtest? Please explain briefly.
28. Any other suggestions?

C Appendix C. AI Declaration

C.1 Declaration

I did knowingly use generative AI tools in this assignment task.

Acknowledgment

In this assignment, I followed the Universitys guidelines for students on academic integrity. No content generated by generative AI tools has been presented as my own work. I take responsibility for the work submitted.

Process: In my assignment preparation, I acknowledge the use of ChatGPT (https://chatgpt.com/) to assist with project architecture planning, code generation, debugging, report polishing, and repository documentation for the AI TRPG project.

Record: I have kept a record of my use of AI tools, including the specific tool(s) used, the prompts submitted, and responses generated.

I understand that my teachers may ask me to provide this information.

C.1.1 Prompt

Report below prompts with a brief description of their usage.

Prompt 1: Help me design the architecture of a web-based AI TRPG built with Next.js. I want the system to include a front-end website, API routes, LLM-based narration, dice checks, state tracking, and scene switching.
Description 1: Used to plan the technical architecture of the project and clarify how the website, backend logic, and AI narrative system should work together.

Prompt 2: Help me define the game state structure for my AI TRPG project. I need fields for role, current scene, danger, HP, clues, inventory, story flags, and session progress.
Description 2: Used to design the main data structure for tracking player status and story progression.

Prompt 3: Help me write the API route for starting a new game session in Next.js, including role selection, scenario selection, and initialization of the game state.
Description 3: Used to generate and revise the backend logic for starting a playable session.

Prompt 4: Help me write the API route that handles player actions in the AI TRPG. It should read the current state, trigger skill checks when necessary, update flags, and return the next narration.
Description 4: Used to build the main interaction logic for the project.

Prompt 5: Help me design a dice-check system for observation, persuasion, and willpower. I want each role to have different modifiers, and I want the result to affect the story state.
Description 5: Used to design the rule-based mechanics that support role-playing and uncertainty in the game.

Prompt 6: Help me add a visible dice-roll animation and calculation display to the front end, so users can understand why a check succeeds or fails.
Description 6: Used to improve the usability and visual clarity of the dice mechanic.

Prompt 7: Help me fix the right sidebar so it stays visible while the page scrolls. I want danger level, character attributes, and roll-related information to remain readable during gameplay.
Description 7: Used to debug and improve the interface layout during gameplay.

Prompt 8: Help me insert generated story images into the correct position in the narrative flow instead of keeping them fixed at the top or bottom of the page.
Description 8: Used to improve image placement and keep visual content aligned with story progression.

Prompt 9: I generated 14 story images in PNG format. Help me revise my code so the website can load the PNG files correctly without changing the rest of the interaction logic.
Description 9: Used to update the image-loading logic after the final image assets were generated.

Prompt 10: Help me fix the story progression logic. The plot currently moves too fast, and I want the whole experience to last around 20–30 minutes with more investigation details and slower pacing.
Description 10: Used to expand and restructure the gameplay flow so the demo would feel more complete.

Prompt 11: Help me rewrite my story into a clearer suspense structure with major phases such as entering the abandoned school, exploring the Student Wellness Center, finding the underground area, revealing the player's identity, and escaping with evidence.
Description 11: Used to organize the authored script into playable stages for the game.

Prompt 12: Help me adjust the story setting so that the player can choose detective, hacker, or priest, and make each role meaningful in the same story world.
Description 12: Used to refine role design and connect character identity with the narrative system.

Prompt 13: Help me debug the page.tsx and route.ts files. I am getting errors in state handling and scene updates, and some features disappear after new changes are added.
Description 13: Used to troubleshoot coding errors during development and integration.

Prompt 14: Help me analyze the questionnaire results from 26 valid responses and identify the strongest findings and the main weaknesses of the AI TRPG demo.
Description 14: Used to interpret user testing results for the report.

Prompt 15: Help me write the Introduction, Why Generative AI is the Right Tool, Approach and Methods, and Conclusion sections in LaTeX for my final report.
Description 15: Used to draft and polish the main written sections of the report.

Prompt 16: Help me shorten my report so it fits the word limit, while keeping the language

natural and reducing repetition.
Description 16: Used to revise the report into a more concise final version.

Prompt 17: Help me write a detailed README for my GitHub repository based on the actual project structure, technical stack, gameplay logic, and deployment setup.
Description 17: Used to create project documentation for the code repository.